

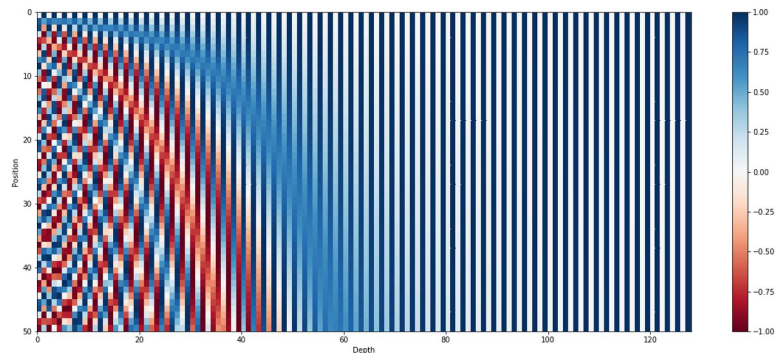
Positional Encodings in Transformers



Positional embeddings

Why positions matter

- Attention is permutation-invariant — without position info, the model sees a bag of words *
- Sinusoidal PE: inject position via sin/cos at different frequencies



ALiBi

Attention with Linear Biases

- Ditch positional embeddings entirely
- Add a linear penalty to attention scores based on relative distance

RoPE

Rotary Position Embedding

- Encode position by rotating embedding dimension pairs
- $\theta_d = 10000^{-2d/D}$
- $R(m, d) = e^{i \cdot m \cdot \theta_d}$
- Dot product between rotated embeddings depends only on relative distance
- Each dimension pair rotates at a different frequency

ABF — Adjusted Base Frequency

- Problem: attention scores decay too rapidly at long distances
- Solution: increase the RoPE base frequency (e.g. 10k $\rightarrow$ 500k or 1M)
- $\theta_k = b'^{-2k/D}$ where $b' \gg 10000$
- Stretches all rotation wavelengths $\rightarrow$ effectively interpolates positions

Training recipe

- First pretrain with small context window
- Then increase base frequency during continued pretraining on long-context data

YaRN — Yet another RoPE extension

Key insight: don't treat all frequencies the same

- High-frequency dims encode local structure → leave them alone
- Low-frequency dims encode global position → apply full interpolation
- Medium frequencies → smooth blend between the two

YaRN Dynamic scaling

- Adjust scaling factor dynamically while generating
- Can be used at inference time without fine-tuning (to some extent)
- Requires caching before RoPE is applied
- Trade-off: works for moderate extensions, degrades at extremes

All together — comparison

1. Standard RoPE

$$\theta^d = 10000^{-2d/D}$$

$$R(m, d) = e^{i \cdot m \cdot \theta_d}$$

2. ABF (adjusted base frequency)

$$\theta^d = b'^{-2d/D} \text{ where } b' \gg 10000$$

Same formula, just replace base 10000 with a larger value (e.g. 500k, 1M)

3. Position interpolation (PI)

$$R(m, d) = e^{i \cdot (m/s) \cdot \theta_d}$$

where $s = L'/L$ (target length / training length)

All positions divided by the same scale factor

RoPE and its context-extension variants: formulas

4. NTK-aware interpolation

$$\theta^d = (b \cdot s^{D/(D-2)})^{-2d/D}$$

Scales base by $s^{D/(D-2)}$ — high-freq dims barely change, low-freq dims get strong interpolation. One parameter, non-uniform effect.

5. YaRN

$$\theta'^d = \theta^d / h(r^d)$$

where $r^d = \text{wavelength}(\theta^d) / L$ (ratio of wavelength to training length)

$h(r)$ is the interpolation ramp:

$$h(r) = 1 \text{ if } r \ll 1 \text{ (high freq } \rightarrow \text{ no change)}$$

$$h(r) = s \text{ if } r \gg 1 \text{ (low freq } \rightarrow \text{ full PI)}$$

$$h(r) = \text{lerp}(1, s) \text{ otherwise (smooth blend)}$$

+ attention temperature: $t = 0.1 \cdot \ln(s) + 1$

NoPE — No Positional Encoding

- Drop all explicit positional encoding
- Causal mask provides implicit position: token m attends to exactly $m+1$ tokens
- BOS trick: attention weight on BOS = $1/(m+1)$ → encodes absolute position
- **Works worse on in-distribution lengths**
- **But generalizes better to unseen lengths!**

RNoPE

- Sometimes apply RoPE, sometimes don't
- Idea: get the best of both worlds
- RoPE layers: strong in-distribution position awareness
- NoPE layers: better length generalization
- Hybrid approach — select which layers get RoPE vs none

DroPE — Drop RoPE

- Just drop RoPE midtraining and it works (*Trust*)
- Remove positional encoding partway through training
- Model retains learned positional awareness from causal mask
- Reportedly improves generalization without sacrificing quality

Thank you

Questions?